

Kris(Qi) Zhou

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Experiences

Industrial Securities Co., Ltd.

Mar. 2025 – Sep. 2025

Software Engineer Intern, Developer Tools & Experimentation Platform

Beijing, China

- Built internal **Java** tooling for AST-based static analysis with JavaParser across a **200K+ LOC** codebase spanning **50+ services**, eliminating most manual code inspection and giving reviewers repeatable verification on every change.
- Redesigned dependency graph traversal on PostgreSQL to support downstream impact analysis before release, reducing analysis time by **2x** and lowering peak memory usage by **40%**.
- Delivered real-time monitoring for **3K+ concurrent A/B experiments** using Kafka and Flink, surfacing rollout metrics with sub-second latency so regressions could be caught quickly.
- Worked with reviewers and platform users to refine analysis rules, failure reports, and monitoring outputs so engineering teams could understand code impact before deployment.

Guotai Haitong Securities

Jan. 2024 – May 2024

Full-Stack Software Engineer Intern, Market Data Platform

Beijing, China

- Built the backend for an equity price monitoring platform serving traders and analysts across multiple regions, ingesting Level 2 market data through Kafka and exposing services with **Spring Boot and Spring Cloud**.
- Developed **React and Redux Toolkit** frontend views for real-time price monitoring, trend visualization, and manual-review replacement workflows.
- Documented **40+ REST APIs** with Swagger/OpenAPI and maintained test coverage with **JUnit 5** and Apache JMeter for core service and performance checks.
- Automated testing, image builds, and Kubernetes deployment through a self-hosted GitLab CE pipeline with dedicated runners.

Institute of Economics, Chinese Academy of Social Sciences

Mar. 2023 – Aug. 2023

Research Assistant, JVM Testing Systems

Beijing, China

- Designed a **Monte Carlo** mutation strategy in **Java** over JVM bytecode execution paths, uncovering defects in production JVM implementations. Developed a coverage guided evolutionary algorithm that steered mutants toward higher branch, code and JVM instruction coverage.
- Fuzzed **HotSpot and OpenJ9** and diffed their execution behavior, surfacing **50+ defects including 3 critical issues** reported upstream to the **IBM OpenJ9** team.
- Organized defect cases with triggering inputs, observed behavior, coverage signals, and reproduction notes for upstream review.

Projects

TeamFlow Hub – Full-Stack Product Collaboration Platform

- Built **TeamFlow Hub**, a full-stack product collaboration app with a **React/TypeScript frontend and Java Spring Boot backend** for managing feature requests, project tasks, user feedback, and release status.
- Designed PostgreSQL tables and REST APIs for users, teams, projects, feature requests, assignments, comments, status history, and audit logs.
- Created responsive UI pages for request submission, task boards, search/filter views, role-based updates, error states, and accessibility-minded form validation.
- Added backend validation, frontend error handling, unit tests, API tests, and structured logs to improve reliability, debugging, and observability across the app.

Skills

Languages: Python, Java, JavaScript, TypeScript, Go, C/C++, SQL, Shell

Frameworks: Django, FastAPI, Spring Cloud, Node.js, Express.js, React.js, Next.js, Vue.js, PyTorch, Hugging Face

Infrastructure & Data: AWS (EKS, EC2, S3), Google Cloud, Docker, Kubernetes, Linux, PostgreSQL, MongoDB, Redis, Elasticsearch, Kafka, Jenkins, GitHub Actions, Git

Education

University of Washington

Sept. 2025 – June 2027

M.S. in Information Management (Specialization in Data Science & AI) (GPA: 3.9/4.0)

Seattle, WA

Peking University

Sept. 2023 – June 2025

Minor in Economics (GPA: 3.8/4.0)

Beijing, China

Beijing International Studies University

Sept. 2021 – June 2025

B.A. in Turkish (GPA: 3.9/4.0)

Beijing, China